

S M Jishanul Islam

AI Engineer, pipikwan pêhtâkwan, Alberta, Canada

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RESEARCH INTERESTS

Multi-modal Learning – Multi-modal Representation – Multi-modal Understanding – Multi-modal AI Applications.

EDUCATION

United International University, Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering (**Major**: Data Science)

Supervisor: Prof. [Dr. Swakkhar Shatabda](#)

2020 — 2024

CGPA: 3.77/4.00

Major CGPA: 4.00/4.00

EXPERIENCE

pipikwan pêhtâkwan

AI Engineer

Alberta, Canada

September 2025 — Present

- Finetuned LLMs catered to multi-purpose tasks for Indigenous Peoples, raising BERTScore performance by 20%.
- Built inference pipelines for billion-parameter models to serve large user bases, reducing response times to around 3 ms.
- Collaborated with the software engineering team to integrate LLM pipelines to build AI-powered sub-products.

Apurba Technologies Ltd.

Software Engineer

Dhaka, Bangladesh

December 2024 — September 2025

- Built scalable and optimized core ML services, improving performance by 10%.
- Co-created the first CLIP-style word recognizer for Bengali, recording an accuracy of 95%+.
- Collaborated with stakeholders to implement, research, and complete project deliverables 2-3 weeks before the deadline.

United International University

Lecturer (contractual)

Dhaka, Bangladesh

June 2024 — October 2024

- Taught core CS concepts to students.
- Took exams to monitor students' performance to provide informative feedback, improving average performance by 7%.
- **Courses conducted:** Object-Oriented Programming Lab, Data Structures and Algorithms 1 Lab, and Database Management Systems Theory.

United International University

Undergraduate Assistant

Dhaka, Bangladesh

September 2023 — May 2024

- Managed to facilitate course content and materials with the lab faculty, checked and evaluated assessments, and judged lab projects.
- Created the first course materials for the country's first-ever undergraduate degree in Data Science.
- **Courses supervised:** Programming for Data Science, and Object-Oriented Programming for Data Science.

PUBLICATIONS

Journals

Under Review

- **Islam, S. M.**, Mustakim, S. H., Hossain, M., Mashira, M., Shourav, N. I., Ahmed, M. R., Islam, S., Shatabda, S., & Islam, A. K. M. "An audio video-based multi-modal fusion approach for speech emotion recognition.", **Knowledge-Based Systems (Elsevier)**, I.F 7.6, under 2nd revision, 2024.

Conference and Workshop papers

- Mustakim, S. H., **Islam, S. M.**, Muna, U. M., Chowdhury, M., Islam, M. J., Ahmmed, S., Sikder, T., Dhruvo, S. T. A., & Shatabda, S. (2025). Watch, Listen, Understand, Mislead: Tri-modal Adversarial Attacks on Short Videos for Content Appropriateness Evaluation. Long-paper (proceedings) track in **SVU Workshop at the IEEE/CVF International Conference on Computer Vision (ICCV)**, 2025. [arxiv.2507.11968](#).
- **Islam, S. M.**, Mustakim, S. H., Ahmmed, S., Sayeedi, M. F. A., Khandoker, S., Dhruvo, S. T. A., & Hossain, N. (2024). MIMIC: Multimodal Islamophobic Meme Identification and Classification. 3rd **Muslims in ML Workshop – NeurIPS**, 2024. [arXiv:2412.00681](#).
- Ahmmed, S., Rahman, T., **Islam, S. M.**, Reyad, A. M., Dey, S., & Farid, D. M. (2024, May). E-MedViTR: Enhanced Vision Transformers with Registers for Biomedical Image Classification. In 2024 6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT) (pp. 664-669). IEEE. [10.1109/ICEE-ICT62016.2024.10534573](#)

Under Review

- Hassan, M., **Islam, S. M.**, Rabby, A. K. M., & Rahman, F. 2025. Beyond Characters: Position-Aware Metric and a Lightweight LLM for Low-Resource Bangla Punctuation Restoration. In 2025 IEEE International Conference on Big Data (BigData). IEEE.

NOTABLE RESEARCH PROJECTS

An audio video-based multi-modal fusion approach for speech emotion recognition.

- **Co-supervised by:** Md. Rayhan Ahmed, Dr. Salekul Islam, Dr. Swakkhar Shatabda, and Dr. A.K.M Muzahidul Islam.
- Under Review at Knowledge-Based Systems (Elsevier) [codebase].
- We present an approach to classify human emotions by fusing audio and visual inputs. Our approach sets new state-of-the-art results while keeping the architecture very simple. Additionally, we present a new frame filtering strategy to overcome the problem of spatiotemporal redundancy.

ChimeraBreak: a novel coordinated tri-modal attack on MLLMs

- **Supervised by:** Dr. Swakkhar Shatabda.
- Long paper (Proceedings), SVU Workshop @ ICCV 2025. [paper(arxiv)] [code] [dataset].
- We introduce SVMA, an adversarial dataset for content moderation in short-form videos, and ChimeraBreak. This coordinated strategy exposes systemic safety flaws in leading MLLMs for content appropriateness evaluation. We experiment with open and closed source MLLMs and utilize LLM-as-a-Judge to evaluate ethical reasoning and confidence.

MIMIC: Multimodal Islamophobic Meme Identification and Classification

- **Supervised by:** Nahid Hossain.
- Accepted at MusIML Workshop - NeurIPS 2024. Ongoing intl. collaboration. [arXiv version] [OpenReview] [codebase].
- First work to combat Islamophobic hate shown through memes by creating a novel dataset out of Islamophobic memes. We also propose a classifier based on the Vision-and-Language Transformer (ViLT) specifically tailored to identify anti-Muslim hate within memes by integrating both visual and textual representations.

NOTABLE PROJECTS

NurtureAid (github.com/sadia-ahmed/Agewell-Frontend)

Built a real-time AI-based cross-platform mobile application that simplifies care through caretakers. Led the app's backend and frontend development as a full-stack engineer.

Tech Stack: React Native, Node.js, Flask, MongoDB, Firebase, LLaMA-Index, PyTorch.

Acknowledgements: Champion of the UIU CSE Project Show, Software Engineering Laboratory; Champion of the Hult Prize OnCampus round in UIU; Selected for the Hult Prize Summit in Boston.

Haven (github.com/S-M-J-I/Haven-Harvard-Hsil-Hackathon-2025)

Created a BCI-powered mental health speech-to-speech conversational agent that understands and responds with emotion. Led the app's backend, frontend, and AI development as a full-stack engineer.

Tech Stack: Python, OpenAI, Emotiv, Firebase, Streamlit.

Acknowledgments: Winner, Harvard HSIL Hackathon Dhaka Hub 2024

BhaShammo (<https://github.com/S-M-J-I/bhashammo-dgt-regional-bengail-2-ipa>)

Solving the unique problem of IPA Transcription of Bengali Regional Dialects using Dialect Guided Tokens.

Tech Stack: Python, Hugging Face, PyTorch.

Acknowledgements: Champion of the Bhashamul national NLP datathon.

SKILLS

Programming: C, C++, Java, Python, JavaScript, PHP, SQL, Solidity.

Machine Learning: PyTorch, Tensorflow, Scikit-Learn, Matplotlib, Numpy, Pandas, NLTK, HuggingFace.

Frameworks/Libraries: React.js, Node.js, Spring Boot, FastAPI, Firebase.

DevOps Tools: Git, GitHub, Docker.

Soft Skills: Communication, Leadership, Design Thinking, Public Speaking, Team Player.

ACHIEVEMENTS

Winner, Harvard Health Systems Innovation Lab Hackathon 2025 (Dhaka Hub)	2025
Academic Excellence Scholarship	2020 — 2024
Gold Award (Champion), International Blockchain Olympiad (IBCOL) 2023	2023
Champion, Bhashamul: Bengali Regional Text to IPA National NLP Datathon	2024
Champion, Hult Prize OnCampus Round	2024
Summit Participant, Hult Prize Summit in Boston, USA	2024
Silver Award(1 st Runner Up), Bangladesh Blockchain Olympiad (BCOLBD) 2023	2023
1 st Runner Up, Intra-University Deep Learning Sprint	2023
Finalist, ICT Innovation Grant (for NurtureAid)	2023
Champion, Software Engineering Lab	2023
Champion, Database Management Systems Lab	2022
Champion, Advanced Object Oriented Programming Lab	2022
The Daily Star Award	2017

EXTRA CURRICULARS

Triumph of IBCOL 2023 and the Future of Blockchain in Bangladesh by Jamuna TV - [interview link](#).

Trainer, Hands-on Workshop on Automatic Speech Recognition by Bengali.AI and IUT Computer Society - [workshop link](#).
Trainer, Hands-on Workshop on building a dApp from scratch by UIU Computer Club - [workshop link](#).

REFERENCES

Dr. Swakkhar Shatabda

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BRAC University, Dhaka, Bangladesh
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Dr. Salekul Islam

Professor of Electrical and Computer Engineering
North South University, Dhaka, Bangladesh
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